LESSON PLAN

LOW SPEED FLIGHT (9)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	11
Total No. of Lesson Plans	13	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Need for simple oxygen system and pressurized cabin effect of lack of O2, Types of O2 regulators Maintenance of O2 system, servicing and safety
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Need for simple oxygen system and pressurized cabin effect of lack of O2, Types of O2 regulators Maintenance of O2 system, servicing and safety precautions.
5. Specific Objectives	9.1.1.1 Trainee will be able to describe various Aircraft engineering systems.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of O2 regulator types; Inspection of maintenance logs; Safety briefing on servicing procedures
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the aircraft oxygen system?
(b) How does lack of oxygen affect human performance at altitude?
(c) Why is a simple oxygen system preferred over complex systems for this course?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Physiological effects of hypoxia • Need for simple oxygen systems • Pressurized cabin requirements • Types of oxygen regulators • Oxygen system maintenance basics	L & Ds	PPT, IB	15 min
(b)	• Servicing procedures for O2 systems • Safety precautions during maintenance • Regulator inspection criteria • Leak detection methods • Storage and handling of oxygen cylinders	L & Ds	PPT, PDF	20 min
(c)	• Common faults in O2 systems • Troubleshooting regulator issues • Documentation and record keeping • Emergency oxygen deployment • Final safety checklist review	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a demand regulator and a continuous flow regulator.
(b) List three critical safety precautions when servicing an oxygen system.
(c) Describe the effect of hypoxia on an aircrew member during flight.

12. Home Assignment and Follow Up	Prepare a brief report on the maintenance schedule for oxygen systems as outlined in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Principle of operation of gas turbine engine.	Refer to the master syllabus reference list	LP 12

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min